

Forecasting Rare Public Operational Actions from Auditable Event Streams

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Abstract

Rare public operational actions are easy to narrate after they occur and difficult to forecast without leaking post-event information. We study a single auditable stream: public announcements of special Codex usage-limit resets. The dataset contains 41 accepted announcements, 315 daily periods, 1,260 six-hour periods, and prediction-safe official incident contexts. Expanding-window historical experiments compare global and adaptive event rates, renewal features, calendar features, and an official-incident model. A 30-day rolling rate is the strongest daily historical baseline (Brier 0.119664), outperforming the calendar model (0.125443); at six-hour resolution the calendar model reaches 0.036443. Official-incident features do not provide stable incremental improvement, and paired block-bootstrap intervals include zero. We therefore freeze a broader baseline set and establish an immutable prospective protocol in which discovery, primary-source verification, prediction, outcome scoring, exclusion, and revision occupy separate append-only layers. Fault drills pass eight predefined failure cases, and an auxiliary audit identifies heterogeneous application outcomes among 20 self-selected public reports, although only one has an exact timestamp. This draft reports historical diagnostics and the prospective design, not prospective superiority. Formal model comparison will occur only after at least 180 valid scheduled days and 20 prospective positives.

1 Introduction

Public organizations sometimes take rare discretionary actions in response to incidents, releases, or salient milestones. Such actions invite retrospective explanations, but a useful scientific question is narrower: can information that was publicly available at time t improve a calibrated forecast of an action after t ? Answering this question requires more than a classifier. The event definition, source hierarchy, observation boundary, missing runs, and later data corrections must all be auditable.

We examine public announcements of special Codex usage-limit resets. The setting is deliberately small and difficult. Events are irregular, the policy regime changes, multiple announcements can refer to one operational action, and reasons are often revealed in the announcement itself. Treating those reasons as predictors would create direct leakage, a general threat to machine-learning science [4]. The project therefore forecasts public organizational behavior rather than inferring a person’s private motives or personality.

Our contributions are fourfold. First, we define an announcement-level gold standard and convert it into daily and six-hour person-period datasets with explicit (start,end] boundaries. Second, we compare global, adaptive-rate, renewal, calendar, and official-incident models under a common expanding-window evaluation. A rolling 30-day rate defeats the initially selected calendar model, illustrating why weak global-rate baselines are insufficient. Third, official-incident features do not improve reliably over temporal baselines. Fourth, we implement a prospective protocol with hash-locked predictions, independent outcome tables, explicit exclusions, missed-run records, and append-only revisions.

The historical experiments are model-development evidence rather than a prospective test. The confirmatory sequence begins only with valid forecasts issued at the frozen daily time. This distinction prevents a favorable historical result from being presented as real-time performance.

2 Related Work

Probabilistic forecast evaluation. The Brier score evaluates squared error for probabilistic outcomes [1]. Strictly proper scoring rules reward honest probabilities rather than thresholded classifications [3]. We use Brier score as the primary metric and log loss as a key secondary metric. Because events are rare, we also report precision–recall area as a descriptive ranking measure, but it does not replace calibration-sensitive scoring.

Out-of-sample and rolling evaluation. Forecast evaluation must preserve temporal order [7]. Random train–test splits would expose a model to future policy regimes and later points in the same event cycle. Leakage can also arise from feature construction, source timestamps, or post-event explanations [4]. Our expanding-window procedure reconstructs features at each issue time and evaluates all models on identical timestamps.

Events and organizational attention. Event system theory characterizes events by novelty, disruption, and criticality [5]. The attention-based view argues that organizational action depends on which issues become salient to decision makers [6]. These theories motivate our context variables, but they do not license causal claims. Official incidents can be associated with both actions and non-actions, and public status records are an incomplete trace of internal attention.

Drift and reflexivity. Changing products and policies create concept drift [2]. In addition, publishing a forecast about a public actor may itself affect discourse or behavior. We therefore preserve model versions, record policy regimes, and forbid silent replacement of historical predictions. The resulting study is closer to an audit-preserving longitudinal diagnostic than to a stationary benchmark.

3 Data and Protocol

3.1 Outcome and observation units

An outcome is an official public statement that a special Codex usage reset is promised, in progress, or completed. Natural personal-window refreshes and unsupported community speculation are excluded. Duplicate posts can map to one action, but the forecast target is the first eligible announcement in a window. The gold set contains 41 announcements and 42 action records because one announcement can encode both hard and banked actions. The main official-account scope is `@thsottiaux` and `@OpenAI`. Promises and completion posts are separate announcements but share an action cluster; one post with two actions yields one announcement outcome and two action rows.

Daily and six-hour rows use the boundary $(s, e]$: an event at s is already known, while an event at e belongs to the future window. Context is snapshotted at s . Official incidents are prediction eligible only when their public status timestamp precedes the snapshot. Reasons first disclosed by a reset announcement remain explanatory metadata and never become predictors of that announcement.

3.2 Models

M0 is the Beta(1, 1) posterior mean of the expanding historical event rate. Stronger rate baselines use rolling 30/60-day windows, a 30-day half-life EWMA, a two-regime rate, a discrete renewal hazard, and nearest-gap analogues. M1 uses $\log(1 + d_t)$, where d_t is time since the latest announcement. M2 adds Pacific-time calendar cycles and a fixed policy-regime indicator. M3-lite combines the renewal term with official incident visibility, active/resolved state, strength, attention, and context age. Logistic models use L2 regularization with $C = 0.25$; continuous variables are standardized inside each training window. The regime boundary is 11 June 2026, the first public banked-reset product evidence. It was selected during historical development rather than before data inspection; M2 without this indicator is therefore a required sensitivity analysis. Changes to the five-hour limit do not define additional regimes.

3.3 Annotation reliability

Historical labels were produced by an LLM rubric followed by human acceptance, not by two independent blinded human coders. A Cohen’s κ or Krippendorff’s α is therefore not identifiable for the original 41 announcements. Future batches use LLM annotation plus independent human review; disagreements are retained, and agreement coefficients will be reported only for new batches with at least 20 candidates. Ambiguous events remain excluded rather than forced into a class.

3.4 Prospective storage

Each scheduled run creates forecasts for global and adaptive-rate baselines, M2, and M3-lite crossed with 24- and 168-hour horizons. A JSON lock stores probabilities, seven projected daily hazards, the issue-time context snapshot, and SHA-256 digests of inputs and code. Outcomes are appended to a separate table only after maturity. Exclusions and revisions have their own tables. Consequently, correcting an outcome cannot rewrite the probability that was issued.

The daily completeness gate uses an independent feed only for candidate discovery. A reset signal must be verified against its original X identifier and accepted into gold before signing. Unknown signals, stale discovery data, validation errors, or duplicate locks create an explicit missed run rather than a backdated forecast.

4 Historical Experiments

4.1 Daily evaluation

The common daily scoring window contains 248 forecasts and 37 positives. Table 1 reports the historical comparison. M2 leads the initial four models, but rolling 30- and 60-day rates both outperform it after stronger baselines are added. Historical M2 is therefore a model-development result, not the strongest available reference.

Table 1: Daily expanding-window results on a common scoring window.

Model	Brier	Log loss	Brier skill vs. M0
M0	0.129385	0.435302	0.000%
M1	0.127290	0.430233	1.620%
M2	0.125443	0.425511	3.047%
M3-lite	0.127800	0.434866	1.226%
Rolling 60d	0.122391	0.411335	5.406%
Rolling 30d	0.119664	0.394584	7.513%
M2 without regime	0.127477	0.433266	1.475%

Scale and calibration. The daily event prevalence is 14.919%, so it is not extremely rare. Absolute daily Brier values are interpreted with prevalence, skill, calibration, and PR-AUC. Rolling 30-day predictions have a calibration intercept of -0.119 and slope of 0.935 , whereas M2 has intercept -0.736 and slope 0.470 .

4.2 Six-hour evaluation

The six-hour dataset has 1,260 periods and exactly 41 positive periods, matching the announcement count. Once the fixed 30-day training requirement and two-positive minimum are applied, 989 forecasts with 38 positives remain. M2 again gives the best point estimates (Table 2), but its paired weekly-block Brier difference from M0 has a 95% interval of $[-0.002073, 0.000235]$. The interval includes zero.

Splitting incident age into 0–6, 6–24, 24–48, and 48–72 hour bins does not improve the renewal baseline. Removing incident strength gives the best context ablation, but its Brier skill over M1 is only 0.080%. These post-development ablations are diagnostic and do not modify the frozen prospective models.

Table 2: Six-hour expanding-window results. PR-AUC is secondary.

Model	Brier	Log loss	PR-AUC
M0	0.037151	0.167388	0.077297
M1	0.037038	0.166331	0.068535
M2	0.036443	0.159763	0.085141
M3-lite	0.037029	0.167232	0.069052

4.3 Bootstrap sensitivity

Paired block intervals are recomputed with 7-, 14-, and 21-day blocks. M2 minus M0 crosses zero for every block length, as does M3-lite minus M2. M2 is worse than rolling 60 days in point estimate, and its difference interval also crosses zero for all three block sizes. With one roughly 315-day sequence, these intervals are sensitivity summaries rather than stable repeated-sample uncertainty.

5 Operational and Application Audits

The automation test suite exercises fresh and stale feeds, missing and unknown signals, duplicate locks, off-schedule issuance, and append-only revision behavior. All eight predefined drills pass. A Windows scheduler runs preflight, forecasting, and scoring at separate times. This arrangement cannot guarantee that external services never fail, but it turns failures into visible logs and missed-run records.

We separately audit whether users report that an announced reset was applied. Among 20 public reports, 13 describe arrival without an explicit mechanism error, four describe arrival with a window or hard-versus-banked mismatch, and three describe failure. Only one report has an exact timestamp; it appears 32 minutes after its announcement. Evidence auditing grades one record A, 14 B, and five C. Reporter identities are not stably captured, so reports from the same discussion cannot be treated as independent Bernoulli observations. These data identify mechanisms and counterexamples, not a population failure rate.

6 Prospective Evaluation

The primary comparison is M2 against the rolling 30-day rate, while M2 against global M0 is retained for compatibility. The theory-feature test asks whether M3-lite improves over M2. Log loss is the key secondary metric; 168-hour scores, calibration, and PR-AUC are secondary analyses. Uncertainty uses paired seven-day block bootstrap with 2,000 resamples.

The first formal comparison requires both 180 valid scheduled days and 20 prospective 24-hour positives. A historical block-variance calculation suggests that reliably detecting a Brier difference of 0.004 may require substantially more than 180 days, so this minimum is not presented as a power guarantee. Bootstrap runs, excluded runs, and missed runs do not count toward the primary sequence. At the time of this draft, no valid scheduled run has matured. The prospective result is therefore intentionally blank. Until the stopping rule is reached, the dashboard reports operational completeness and descriptive cumulative scores without declaring a winner.

7 Limitations, Ethics, and Conclusion

This study concerns one small public event stream. Windows overlap, policies change, official status records can omit local failures, and community application reports are self-selected. The discovery layer also depends on an external service, although final labels require primary evidence. Historical point estimates should therefore not be generalized to other organizations or interpreted causally.

The design avoids claims about private motives, personality, or mental state. It uses public professional actions, discourages attempts to manipulate the observed signals, and records possible reflexive intervention. Rarity also depends on scale: the six-hour event rate is about 3.8%, while the daily rate is about 15%. The

central methodological result is modest: an operational-action forecast can be made auditable by separating discovery, labels, features, locked predictions, outcomes, and revisions. A recent-rate baseline outperforms the selected historical calendar model, and the tested incident features add no stable value. Only the amended prospective sequence can determine whether those orderings persist.

References

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A Appendix: Reproducibility Details

The repository retains raw source indices, bilingual evidence summaries, annotation instructions, adjudication records, daily and six-hour person-period tables, context snapshots, rolling forecasts, locked prospective manifests, automation logs, exclusions, and revisions. All timestamps use UTC. The frozen protocol has SHA-256 `de216beb5dbbf1dbc77ae863f2628b1d750c55f9c7e52f6efed93b0b39d949cb`.

The historical six-hour comparison starts only after 120 valid prior periods and at least two examples of each class. Binary features pass through preprocessing; continuous and cyclic features are standardized using training-window statistics. The paired bootstrap resamples contiguous seven-day blocks to preserve short-range dependence.